

RIDGE LEVERAGE SCORES FOR LOW-RANK MATRIX APPROXIMATION

Michael B. Cohen, Cameron Musco, Christopher Musco

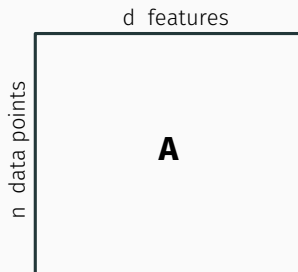
Massachusetts Institute of Technology

“Ridge Leverage Scores for Low-Approximation” =
“Dimensionality Reduction for k-Means Clustering and
Low-Rank Approximation”
+
“Uniform Sampling for Matrix Approximation”

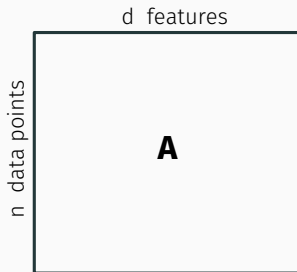
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Papers and slides available at chrismusco.com.

HOW TO DEAL WITH HUGE DATA SETS?

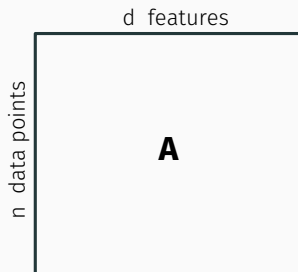


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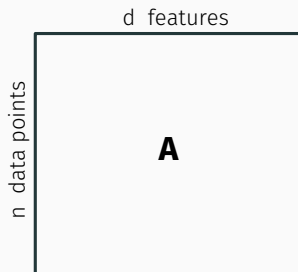
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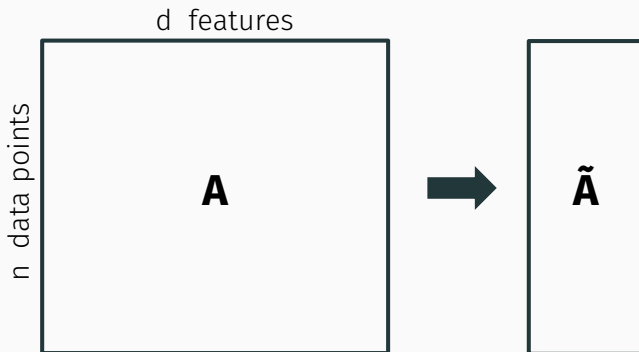
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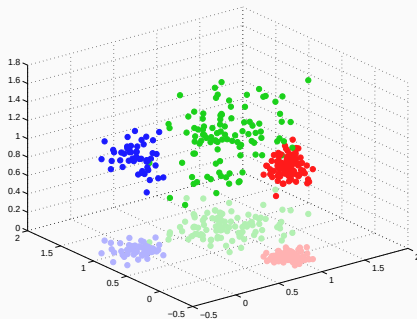


- computing power (MapReduce/Hadoop, Apache Spark, etc.)
- limited data access (iterative methods, stochastic methods)
- dimensionality reduction (“sketch-and-solve”)

Replace high dimensional data with low dimensional **sketch**.

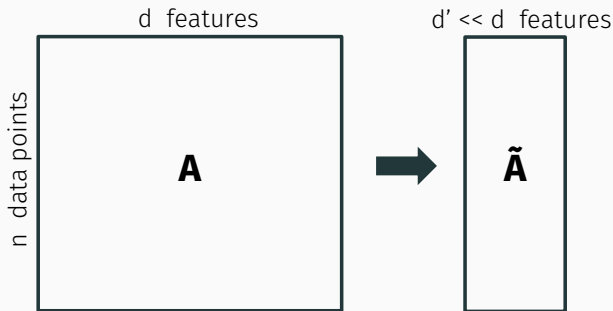


Solution on sketch $\tilde{\mathbf{A}}$ should approximate original solution.



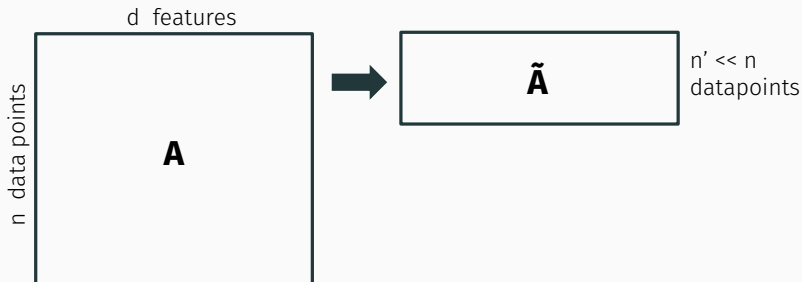
DIMENSIONALITY REDUCTION

Reduce dimensionality of data points, not their number.



DIMENSIONALITY REDUCTION (THE OTHER DIRECTION)

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$\tilde{A}$ is often called a **coreset**.

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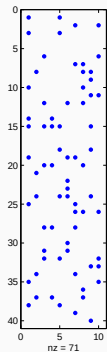
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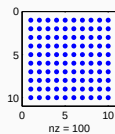
- **Johnson-Lindenstrauss projections** = super fast to apply, naturally adapts to streaming/distributed environments.
- **Deterministic methods (SVD, Frequent Directions)** = best data compression.
- **Data Selection/Sampling** = preserves structure and sparsity.

SKETCHING BY SAMPLING

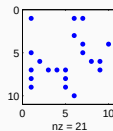
Original Data



General Sketch

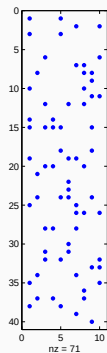


Data Sample

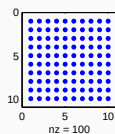


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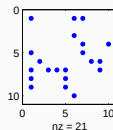
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Sampling is also closely tied to understanding heuristic methods and has produced valuable theory.

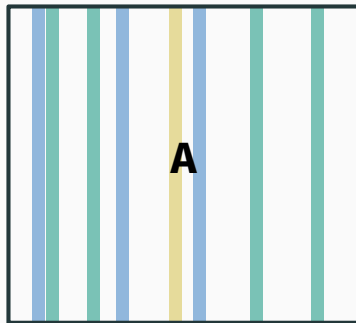
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IMPORTANCE SAMPLING

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1. **Leverage Scores** are used ubiquitously as importance sampling probabilities for matrix sketching.
2. These scores have been extended to sketches for low-rank approximation problems, but **not in a satisfying way**.
3. We give a more natural extension, via **Ridge Leverage Scores**. These scores lead to simple proofs and have a bunch of desirable properties and new applications.

Definition (Subspace Embedding)

A sketch $\tilde{\mathbf{A}}$ such that, for all vectors $\mathbf{x}$, $\|\mathbf{x}^T \tilde{\mathbf{A}}\| = (1 \pm \epsilon) \|\mathbf{x}^T \mathbf{A}\|$.

$$\left\| \begin{array}{|c|} \hline \mathbf{x}^T \\ \hline \end{array} \begin{array}{|c|} \hline \tilde{\mathbf{A}} \\ \hline \end{array} \right\|_2^2 = (1 \pm \epsilon) \left\| \begin{array}{|c|} \hline \mathbf{x}^T \\ \hline \end{array} \begin{array}{|c|} \hline \mathbf{A} \\ \hline \end{array} \right\|_2^2$$

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- Spectral sparsifiers for fast approximate graph algorithms.

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Let's think about subspace embeddings as approximating the quadratic form $\mathbf{A} \mathbf{A}^T$.

The diagram illustrates the quadratic form sampling equation $A A^T = A A^T$. It features three rectangular boxes: a horizontal box on the left containing the symbol A , a vertical box in the middle containing the symbol A^T , and a square box on the right containing the symbol AA^T . An equals sign is positioned between the vertical A^T box and the square AA^T box. To the right of the AA^T box is an empty square box, likely intended for a diagram or explanation of the matrix dimensions.

$$A A^T = A A^T$$

QUADRATIC FORM SAMPLING

The diagram illustrates the process of sampling a quadratic form from a matrix A . It shows the matrix A with its first column highlighted in blue and labeled a_1 . This column is multiplied by the first row of the transpose matrix A^T , which is also highlighted in blue and labeled a_1^T . The result is a blue square representing the outer product $a_1 a_1^T$, which is equal to the first element of the matrix product AA^T .

$$A \begin{matrix} a_1 \\ \vdots \end{matrix} = \begin{matrix} a_1^T \\ \vdots \end{matrix} = a_1 a_1^T = AA^T$$

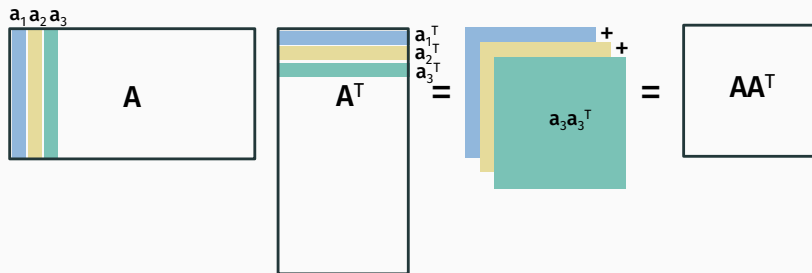
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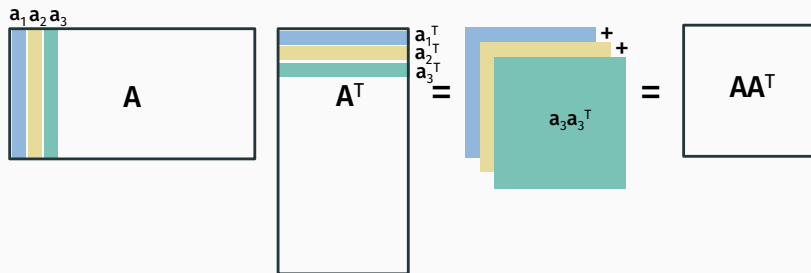
- A matrix A (represented as a rectangle) is multiplied by a vector $a_1 a_2$ (represented as a vertical bar with a blue and yellow segment).
- The result is a vector A^T (represented as a vertical rectangle).
- This vector A^T is then multiplied by a matrix $a_2 a_2^T$ (represented as a square with a blue and yellow segment).
- The final result is the matrix AA^T (represented as a square).

The diagram uses color coding to highlight the dimensions and the resulting quadratic form. The blue and yellow segments represent the sampled components of the vectors and matrices.

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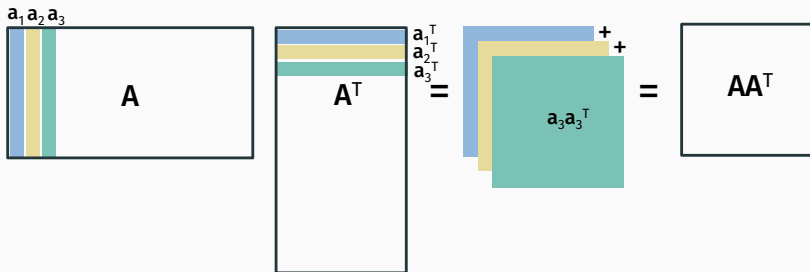


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$$\tilde{A}\tilde{A}^T = \sum_{j \in [1, \dots, d]} w_j a_j a_j^T$$

Sampling Scheme: For any set of sampling probabilities $p_1, p_2, \dots, p_d$ include column $\mathbf{a}_i$ in $\tilde{\mathbf{A}}$ with probability p_i and reweight the column by $\frac{1}{p_i}$.

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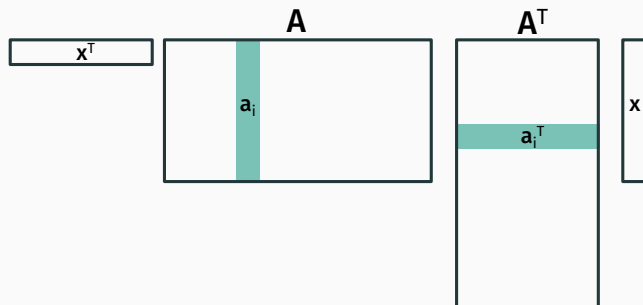
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Need to select more “unique” columns with higher probability.

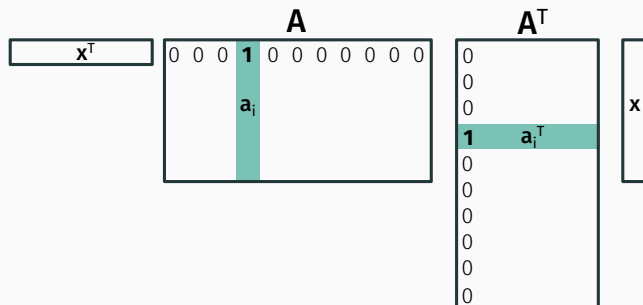
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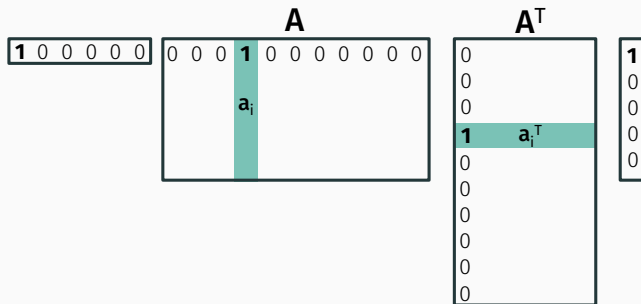
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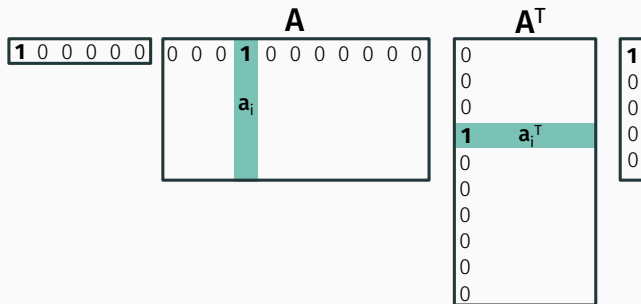
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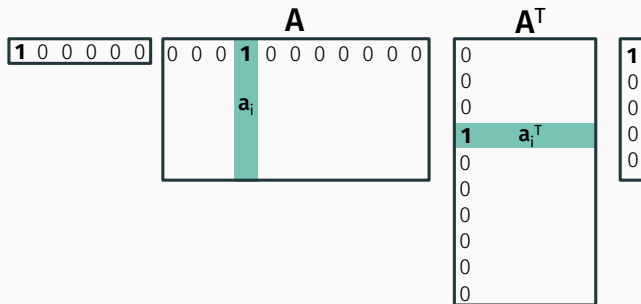
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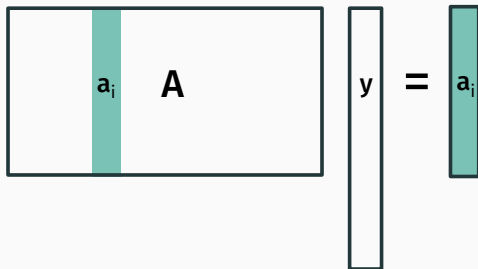
$\mathbf{x}^T \tilde{\mathbf{A}} \tilde{\mathbf{A}}^T \mathbf{x}$ cannot equal $(1 \pm \epsilon) \mathbf{x}^T \mathbf{A} \mathbf{A}^T \mathbf{x}$.

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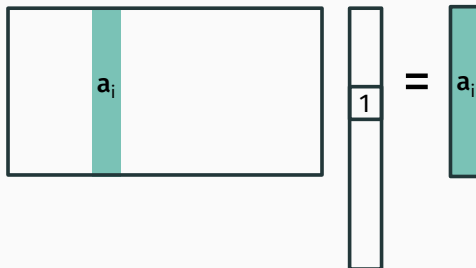
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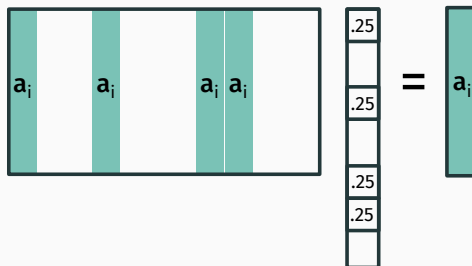


$\tau(\mathbf{a}_i) \leq 1$ since we can choose $\mathbf{y}$ to be the i^{th} basis vector.

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If more columns align with $\mathbf{a}_i$, $\tau(\mathbf{a}_i)$ decreases.

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More specifically, to get a subspace embedding, we sample each column $\mathbf{a}_i$ with probability $\tau(\mathbf{a}_i) \cdot \frac{\log n}{\epsilon^2}$.

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We're approximating $\mathbf{A}\mathbf{A}^T$ with a sum of (binary) random matrices:

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$\tau(\mathbf{a}_i) \frac{\log n}{\epsilon^2}$ is the lowest p_i which ensures $\mathbf{X}_i \preceq \frac{\epsilon^2}{\log n} \mathbf{A}\mathbf{A}^T$.

FINAL SUBSPACE EMBEDDING THEOREM

$$\begin{array}{c} \text{---} \mathbf{x}^T \text{---} \\ \text{---} \mathbf{\tilde{A}} \text{---} \\ \underbrace{\hspace{1.5cm}}_{\tilde{O}(n)} \end{array} \begin{array}{c} 2 \\ 2 \end{array} = (1 \pm \epsilon) \begin{array}{c} \text{---} \mathbf{x}^T \text{---} \\ \text{---} \mathbf{A} \text{---} \end{array} \begin{array}{c} 2 \\ 2 \end{array}$$

Theorem (Subspace Embedding via Sampling)

Sampling $O\left(\frac{n \log n}{\epsilon^2}\right)$ columns from $\mathbf{A}$ by *leverage score* gives an ϵ factor subspace embedding with high probability.

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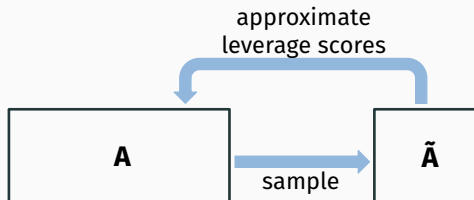
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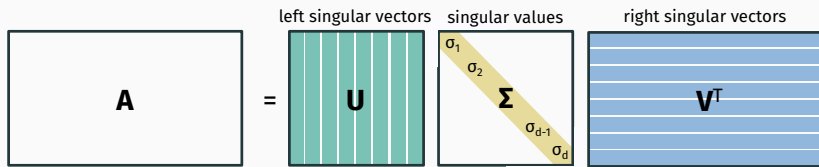
linear system solving, low-rank approximation, k-means clustering, convex optimization, linear programming, matrix completion, multi-label classification, spectral graph problems

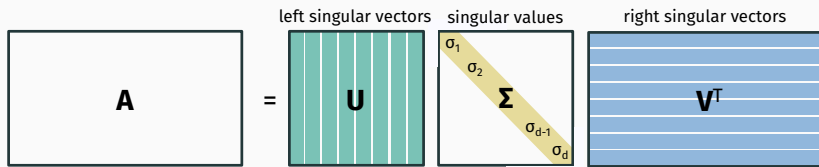
Leverage scores have been very influential, even beyond direct application to subspace embeddings.

linear system solving, low-rank approximation, k-means clustering, convex optimization, linear programming, matrix completion, multi-label classification, spectral graph problems

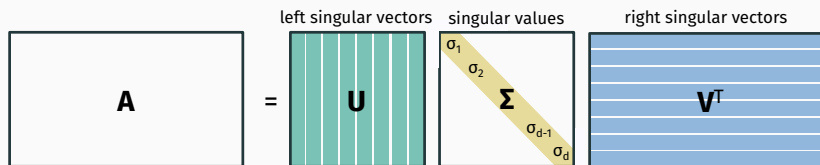
There are many generalizations and modifications of leverage scores.

Extensions to **low-rank** problems have been especially popular.



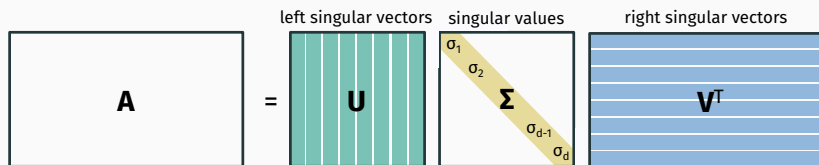


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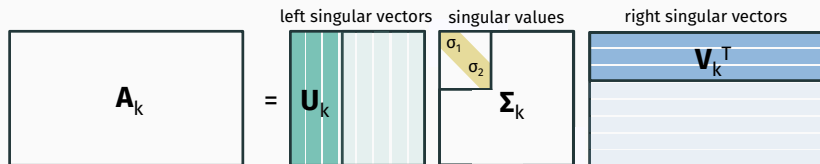
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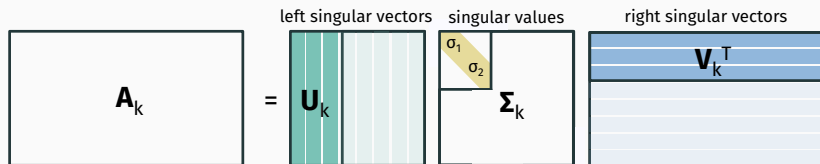


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For $x^T \tilde{A} \tilde{A}^T x \approx x^T A A^T x$ for all x we need to preserve information about every singular direction/value. Specifically, it can be shown that $\sigma_i(\tilde{A}) = (1 \pm \epsilon) \sigma_i(A)$



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In these cases, we should be able to obtain smaller sketches – i.e. $O(k)$ instead of $O(n)$.

Find low-rank matrix close to **A**.

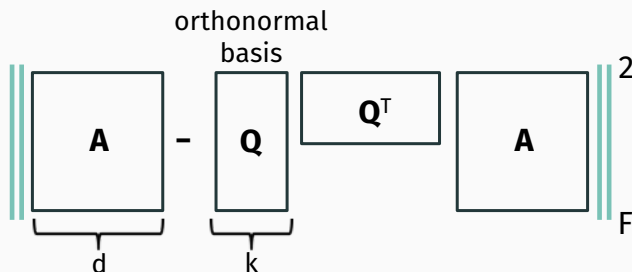
Find low-rank matrix close in Frobenius norm to $\mathbf{A}$.

The diagram illustrates the Frobenius norm distance between a matrix $\mathbf{A}$ and its rank- k approximation $\mathbf{A}_k$. It features two square boxes, one labeled $\mathbf{A}$ on the left and one labeled $\mathbf{A}_k$ on the right, with a minus sign between them. To the left of the $\mathbf{A}$ box are three vertical teal lines. To the right of the $\mathbf{A}_k$ box are three vertical teal lines. A superscript 2 is positioned above the rightmost teal line, and a subscript F is positioned below it, indicating the Frobenius norm squared.

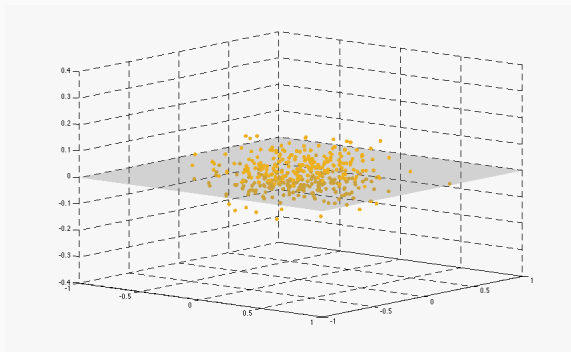
$$\|\mathbf{A} - \mathbf{A}_k\|_F^2$$

rank k

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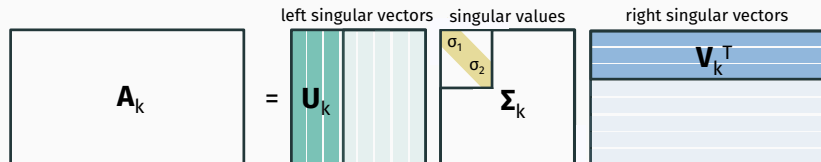
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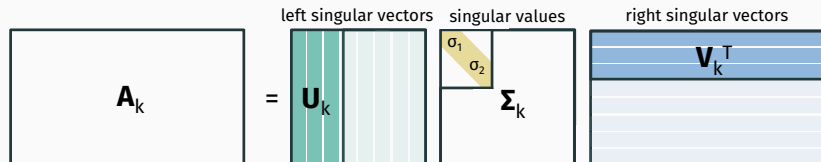
$\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^T\mathbf{A}\|_F^2 =$ sum of squared distances to hyperplane
spanned by $\mathbf{Q}$.

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$$\|\mathbf{A} - \mathbf{A}_k\|_F^2 = \|\mathbf{A} - \mathbf{U}_k \mathbf{U}_k^T \mathbf{A}\|_F^2 = \min \|\mathbf{A} - \mathbf{Q} \mathbf{Q}^T \mathbf{A}\|_F^2.$$

Set $\mathbf{Q} = \mathbf{U}_k$, i.e. to the top k singular vectors of $\mathbf{A}$.

With constraints, Frobenius norm low-rank approximation captures a variety of additional interesting problems:

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- nonnegative PCA
- sparse PCA
- **k-means clustering** (see slides on my website)

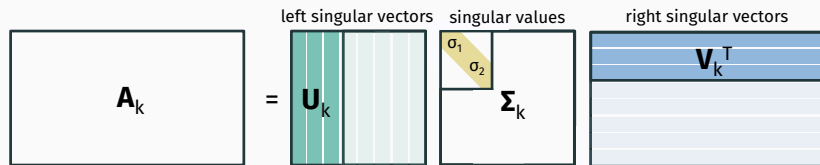
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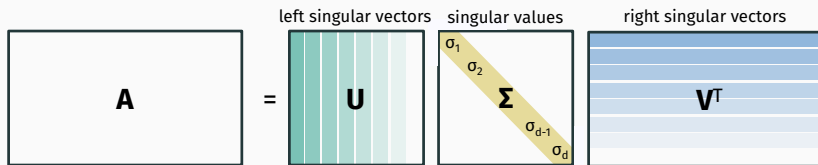
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Column Subset Selection:

Find an $\tilde{\mathbf{A}}$ such that $\|\mathbf{A} - \text{proj}_{\tilde{\mathbf{A}}}(\mathbf{A})\|_F^2 \leq (1 + \epsilon)\|\mathbf{A} - \mathbf{A}_k\|_F^2$.

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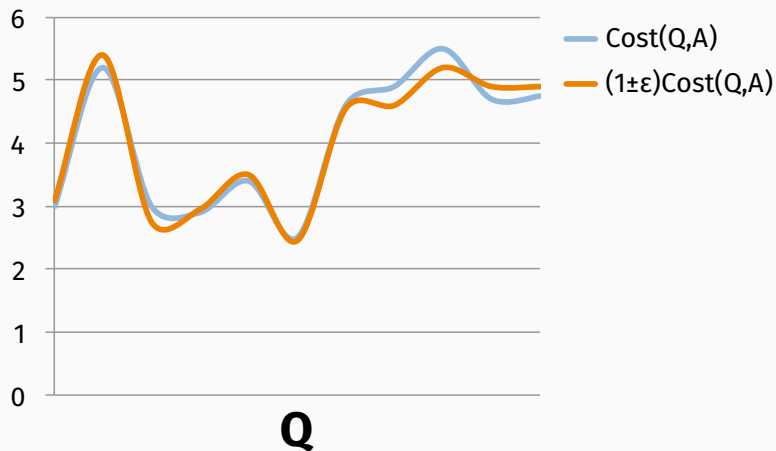
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PROJECTION COST PRESERVATION



$$\|\tilde{A} - QQ^T \tilde{A}\|_F^2 = (1 \pm \epsilon) \|A - QQ^T A\|_F^2$$

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But we would get a sketch with too many samples:
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[Drineas, Mahoney, Muthukrishnan '08, and Sarlós '06]

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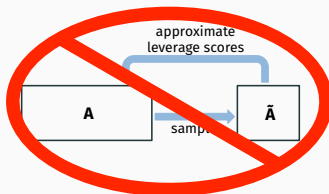
[Cohen, Elder, Musco, Musco, Persu '15]

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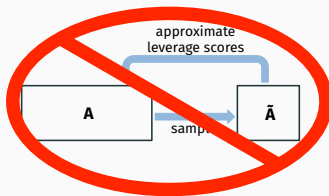
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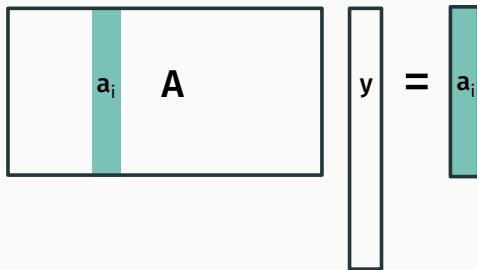
2. The scores cannot be computed in a data stream.

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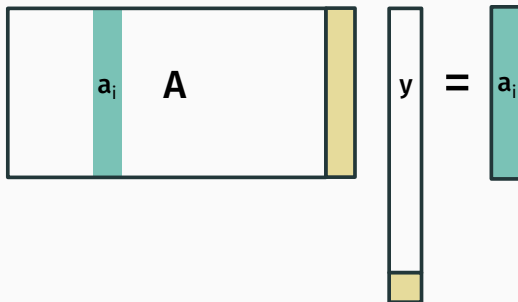
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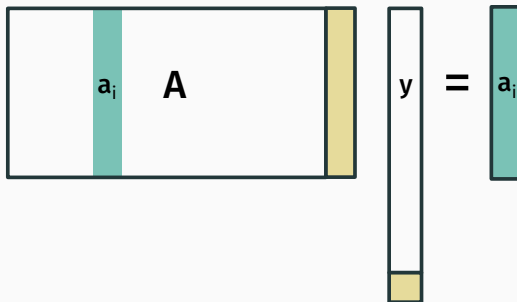
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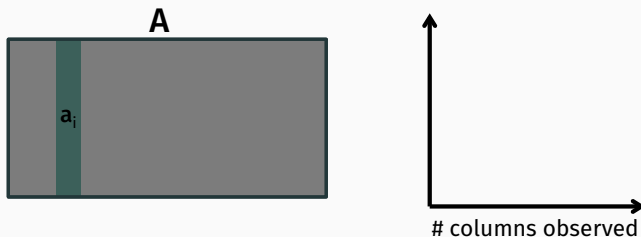
For standard leverage scores, adding a column to A can only **decrease** the importance of existing columns.

Streaming setup:

Receive columns of $\mathbf{A}$ one-by-one. Reject each with probability depending on its (low-rank) leverage score with respect to the columns seen so far [Kelner, Levin '11].

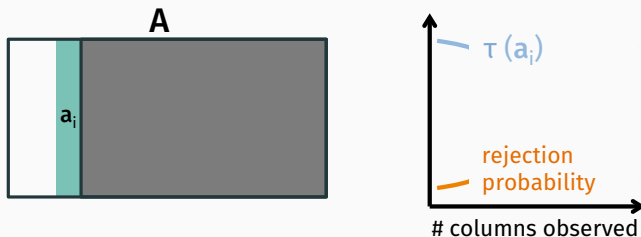
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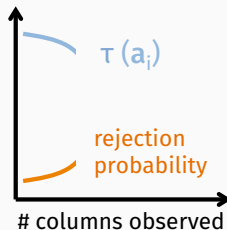
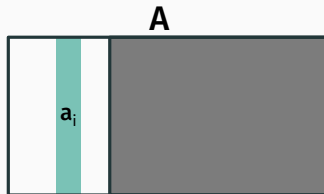
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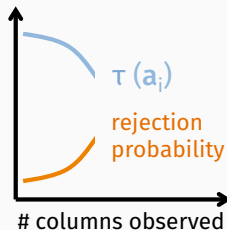
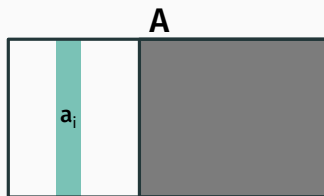
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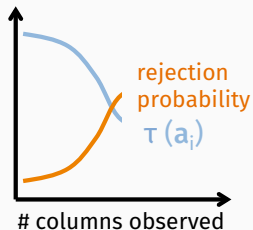
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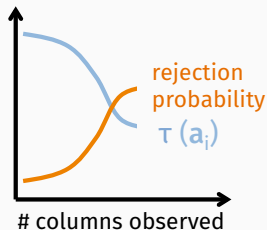
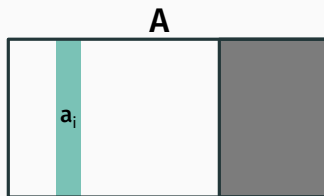
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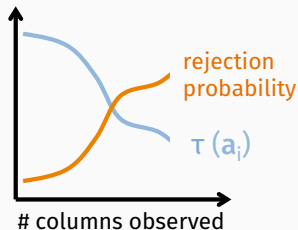
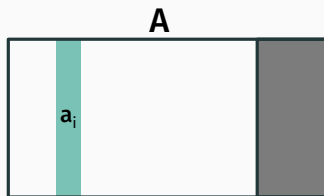
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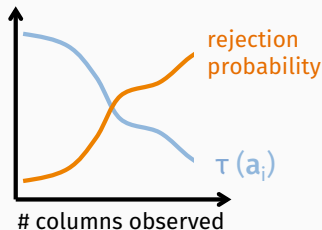
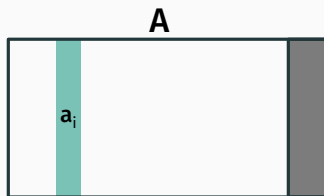
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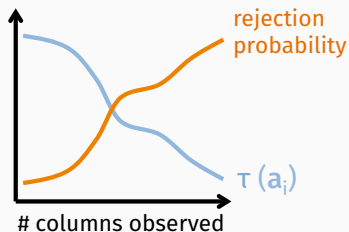
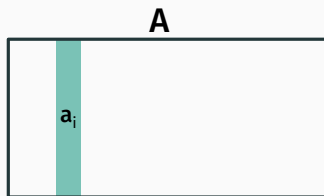
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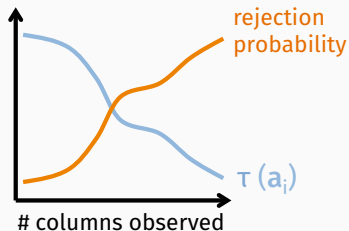
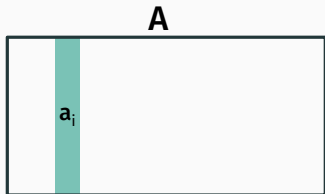
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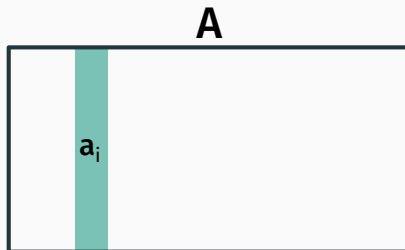
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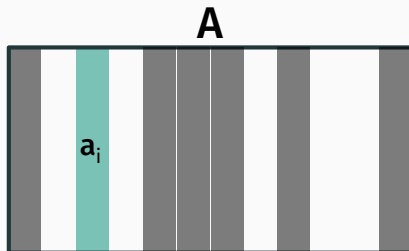


Rejection probability only increase, so we never delete a column with too high of probability.

Iterative Leverage Score Sampling: Monotonicity is essential because it ensures that a **uniform subsample** of columns can at least be used to find upper bounds for leverage scores.
[Cohen, Lee, Musco, Musco, Peng, Sidford '15]



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A														$\mathbf{u}_1$	
1	1	1	1	1	1	1	1	0	0	0	0	0	0	1	
0	0	0	0	0	0	0	0	1	1	1	1	1	1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

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1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	1	1	1	1	1	1	2	1
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

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0	0	0	0	0	0	0	0	1	1	1	1	1	1	1
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Adding a column could cause $\mathbf{a}_i^T (\mathbf{A}_k \mathbf{A}_k)^{-1} \mathbf{a}_i$ to drop significantly.

Here $\mathbf{a}_1^T (\mathbf{A}_1 \mathbf{A}_1)^{-1} \mathbf{a}_1 \Rightarrow 0$.

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“Soften” the existing definition of rank k leverage scores.

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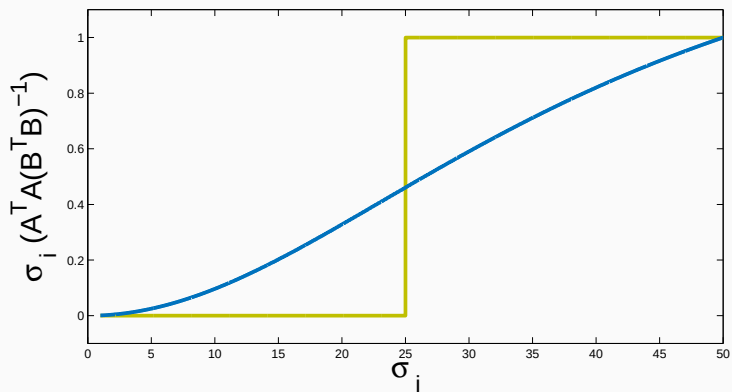
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Relatively “gentle” soft step:



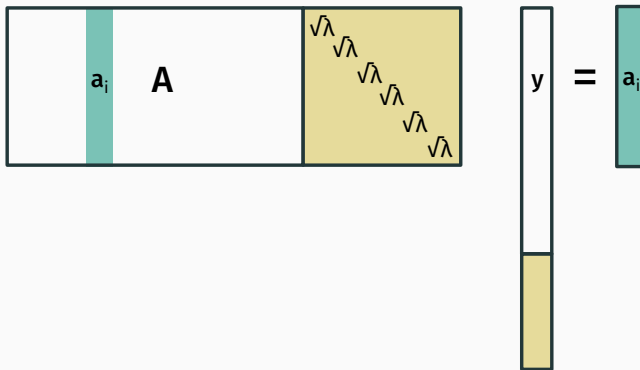
We “wash out” the importance of columns by computing leverage scores over $\mathbf{A}$ with an identity appended:

The diagram illustrates the process of computing leverage scores by appending an identity matrix to matrix $\mathbf{A}$. On the left, a large rectangle represents the matrix $\mathbf{A}$. A vertical teal bar within this rectangle is labeled a_i , representing a column of $\mathbf{A}$. To the right of this rectangle is a tall, narrow white rectangle labeled y , representing an identity matrix. An equals sign follows, leading to another tall, narrow teal rectangle labeled a_i , representing the resulting leverage scores.

$$\begin{bmatrix} \text{Identity} & \mathbf{A} \end{bmatrix} y = a_i$$

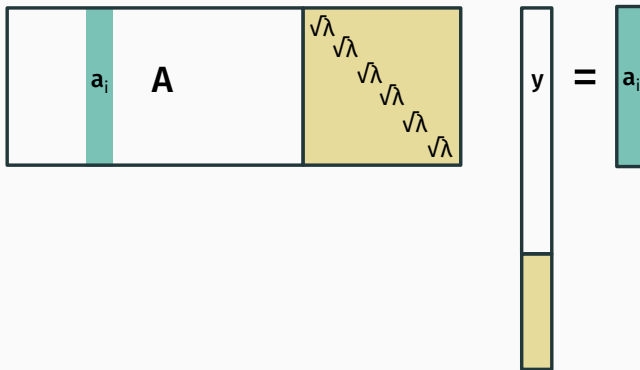
ANOTHER PERSPECTIVE

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Effect is weaker when $\mathbf{a}_i$ aligns with large singular vectors of $\mathbf{A}$.

Theorem (Ridge Leverage Score Sampling)

With λ set to $\|\mathbf{A} - \mathbf{A}_k\|_F^2/k$, sampling $O(k \log k/\epsilon^2)$ columns by ridge leverage score produces an ϵ error projection cost preserving sketch with high probability.

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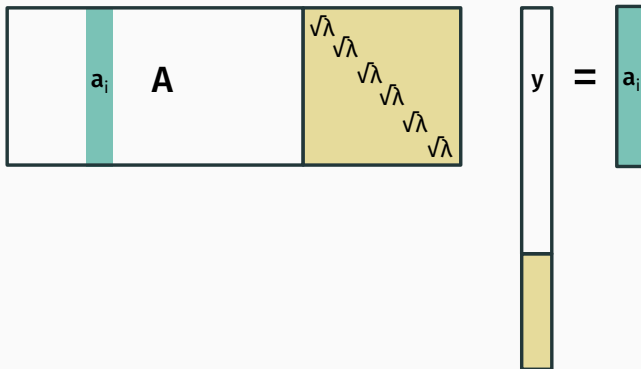
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Furthermore, $(\|\mathbf{A} - \mathbf{A}_k\|_F^2/k)$ -ridge leverage scores are monotonic with respect to column additions.

MONOTONICITY OF RIDGE LEVERAGE SCORES



Since $\lambda = \|\mathbf{A} - \mathbf{A}_k\|_F^2/k$ can only increase as columns are added to $\mathbf{A}$, this perspective immediately implies that ridge leverage scores are monotonic.

With λ set to $\|\mathbf{A} - \mathbf{A}_k\|_F^2/k$, sampling by ridge leverage score produces a sketch $\tilde{\mathbf{A}}$ such that:

$$(1 - \epsilon)\tilde{\mathbf{A}}\tilde{\mathbf{A}}^T - \epsilon\frac{\|\mathbf{A} - \mathbf{A}_k\|_F^2}{k}\mathbf{I} \preceq \mathbf{A}\mathbf{A}^T \preceq (1 + \epsilon)\tilde{\mathbf{A}}\tilde{\mathbf{A}}^T + \epsilon\frac{\|\mathbf{A} - \mathbf{A}_k\|_F^2}{k}\mathbf{I}$$

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Multiplicative error of a **subspace embedding**.

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Both are known to give projection cost preserving sketches.
Handling both errors simultaneously is tedious, but not hard.

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Sum of vector products with $\tilde{\mathbf{A}}$. Each preserved to within a $(1 \pm \epsilon)$ factor, so the entire sum is as well.

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Intuition:

Dealing with rank k operators ($\mathbf{Q}$ is rank k), so we only pay the additive error k times.

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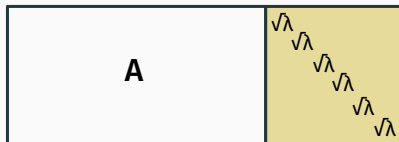
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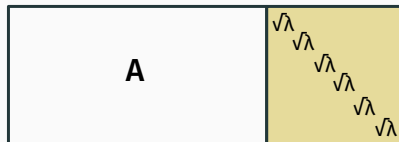
Since $\mathbf{A}_k$ is a better low-rank approximation than any $\mathbf{Q} \mathbf{Q}^T \mathbf{A}$.

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Proof follows directly from our “appending an identity” view!

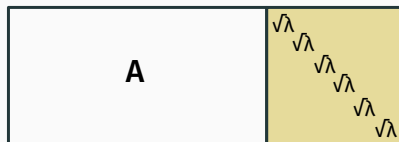


Proof:



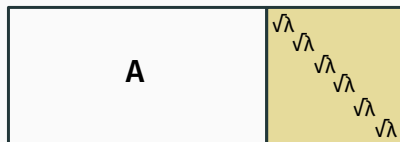
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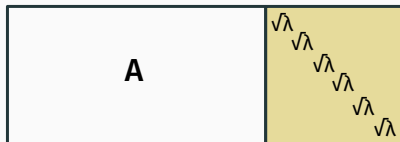
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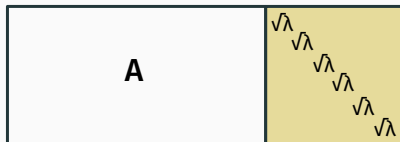
$$(1 - \epsilon)\mathbf{B}\mathbf{B}^T \preceq \mathbf{A}\mathbf{A}^T + \lambda\mathbf{I} \preceq (1 + \epsilon)\mathbf{B}\mathbf{B}^T$$



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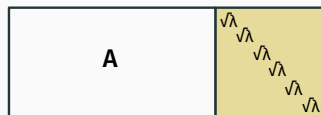
$$(1 - \epsilon)\mathbf{B}\mathbf{B}^T \preceq \mathbf{A}\mathbf{A}^T + \lambda\mathbf{I} \preceq (1 + \epsilon)\mathbf{B}\mathbf{B}^T$$
4. $(1 - \epsilon)(\tilde{\mathbf{A}}\tilde{\mathbf{A}}^T + \lambda\mathbf{I}) \preceq \mathbf{A}\mathbf{A}^T + \lambda\mathbf{I} \preceq (1 + \epsilon)(\tilde{\mathbf{A}}\tilde{\mathbf{A}}^T + \lambda\mathbf{I})$



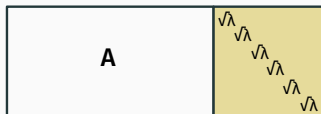
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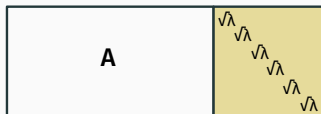


Number of columns sampled to form $\tilde{A}$ depends on sum of leverage scores, **outside of the identity columns**.



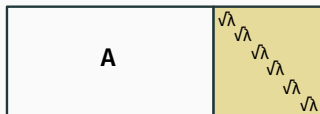
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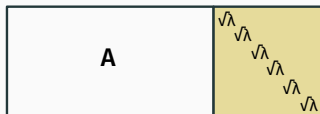
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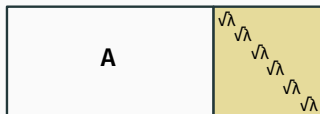
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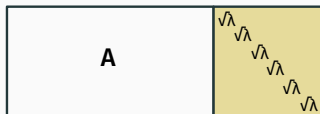
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 &= \sum_{i=1}^d \frac{\sigma_i(\mathbf{A})}{\sigma_i(\mathbf{A}) + \frac{\|\mathbf{A} - \mathbf{A}_k\|_F^2}{k}} \\
 &\leq k + \sum_{i=k+1}^d \frac{\sigma_i(\mathbf{A})}{\sigma_i(\mathbf{A}) + \frac{\|\mathbf{A} - \mathbf{A}_k\|_F^2}{k}}
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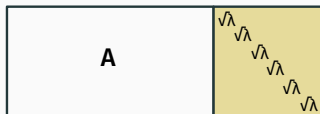
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$$\begin{aligned}
 \sum_{i=1}^d \tilde{\tau}(a_i) &= \text{tr} \left(A^T \left(A^T A + \frac{\|A - A_k\|_F^2}{k} I \right)^{-1} A \right) \\
 &= \sum_{i=1}^d \frac{\sigma_i(A)}{\sigma_i(A) + \frac{\|A - A_k\|_F^2}{k}} \\
 &\leq k + \sum_{i=k+1}^d \frac{\sigma_i(A)}{\frac{\|A - A_k\|_F^2}{k}} \\
 &= k + k = O(k).
 \end{aligned}$$

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Please checkout the arXiv preprint!